



PASSIVE COOLING SYSTEM FOR HOME



A PROJECT REPORT

Submitted by

MANIKANDAN K	(611223114056)
MOHAN G	(611223114060)
NAVEENKUMAR R	(611223114066)
SABARI K	(611223114080)

In partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

**KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM
(An Autonomous Institution)**

ANNA UNIVERSITY : CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report “**PASSIVE COOLING FOR HOME**” is the bonafide work of “**MANIKANDAN K, MOHAN G, NAVEENKUMAR R, SABARI K**” who carried out the project work under my supervision.

SIGNATURE

Dr. K.S.PRABHAKARAN, M.E., Ph.D., MISTE,

HEAD OF THE DEPARTMENT

Associate Professor

Department of Mechanical Engineering

Knowledge Institute of Technology

Salem – 637 504.

SIGNATURE

Mr. S.SANTHOSH, B.E., M.E.

SUPERVISOR

Assistant Professor

Department of Mechanical Engineering

Knowledge Institute of Technology

Salem – 637 504.

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ABSTRACT

The Passive Cooling System for Home is designed to reduce indoor temperature without the use of conventional air-conditioning systems. The system utilizes natural cooling techniques such as proper ventilation, shading, and thermal insulation to maintain a comfortable indoor environment while minimizing energy consumption. The project aims to provide an eco-friendly and cost-effective solution for residential cooling. Temperature data is monitored and analyzed to evaluate the cooling performance of the system. The results show that passive cooling techniques can effectively lower indoor temperatures, improve thermal comfort, and reduce dependence on electrical cooling devices, thereby promoting energy conservation and environmental sustainability.

Keywords: Passive Cooling, Temperature Monitoring, ESP32, Data Logging, Energy Efficiency, Sustainable Architecture

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LIST OF ABBREVIATIONS

RTC	Real-Time Clock
ESP32	Embedded System with Wi-Fi & Bluetooth Microcontroller
IoT	Internet of Things
GPIO	General Purpose Input/Output
SD	Secure Digital
SPI	Serial Peripheral Interface
CSV	Comma Separated Values
ADC	Analog to Digital Converter
HVAC	Heating, Ventilation, and Air Conditioning

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF PASSIVE COOLING

The increasing demand for cooling in residential buildings has led to higher energy consumption and electricity costs. Conventional cooling systems such as air conditioners require a significant amount of electrical power and contribute to environmental pollution. Therefore, there is a growing need for sustainable and energy-efficient cooling solutions.

Passive cooling is a natural method of reducing indoor temperature without using mechanical cooling equipment. It utilizes techniques such as natural ventilation, shading, thermal insulation, and reflective surfaces to maintain indoor comfort. These methods help reduce energy consumption, lower operating costs, and promote environmentally friendly living.

1.2 NEED FOR PASSIVE COOLING

In many regions, especially during summer, indoor temperatures become uncomfortable, leading to increased use of air conditioners and fans. This results in high electricity consumption and increased energy costs. In addition, excessive use of conventional cooling systems contributes to environmental issues such as greenhouse gas emissions.

Passive cooling provides an effective alternative by utilizing natural cooling methods to reduce indoor temperature. It helps improve thermal comfort, conserve energy, reduce operating costs, and support sustainable building practices. Therefore, the implementation of passive cooling systems has become essential for creating energy-efficient and environmentally friendly homes.

1.3 PRINCIPLES OF PASSIVE COOLING

Passive cooling works by reducing heat gain and improving natural heat removal from a building. It utilizes natural elements such as airflow, shading, and thermal insulation to maintain a comfortable indoor environment. Proper ventilation allows fresh air to circulate through the building, while shading devices reduce direct solar heat entering the house.

The effectiveness of passive cooling depends on factors such as building orientation, material selection, window placement, and climatic conditions. By combining these techniques, indoor temperatures can be reduced without the need for mechanical cooling systems, resulting in energy savings and improved thermal comfort.

1.4 LIMITATIONS OF CONVENTIONAL COOLING SYSTEMS

Conventional cooling systems such as air conditioners and air coolers require a significant amount of electrical energy for operation. Continuous usage increases electricity consumption and results in higher operating costs. In many regions, the growing demand for cooling places additional pressure on power generation and distribution systems.

These systems also contribute to environmental concerns through increased energy usage and greenhouse gas emissions. Regular maintenance, installation costs, and dependence on electricity make conventional cooling methods less sustainable. Therefore, there is a need for alternative cooling solutions that can provide thermal comfort while minimizing energy consumption.

1.5 IMPACT OF EXCESSIVE ENERGY CONSUMPTION

The excessive use of conventional cooling systems has a significant impact on energy consumption and the environment. Air conditioners and other mechanical cooling devices

require large amounts of electricity, leading to increased energy demand and higher utility costs. As the number of cooling appliances continues to grow, the strain on power generation systems also increases.

In addition, the extensive use of electrical cooling equipment contributes to greenhouse gas emissions and environmental degradation. Reducing energy consumption through sustainable cooling methods is essential for conserving natural resources and promoting environmentally friendly living practices.

1.6 NEED FOR PASSIVE COOLING SYSTEMS

To overcome the limitations of conventional cooling methods, there is a growing need for passive cooling systems in residential buildings. Passive cooling utilizes natural techniques such as ventilation, shading, thermal insulation, and reflective materials to reduce indoor temperature without the use of mechanical cooling equipment.

These systems help minimize energy consumption, lower electricity costs, and reduce environmental impact. By providing a comfortable indoor environment through sustainable design strategies, passive cooling systems contribute to energy conservation and support the development of eco-friendly and energy-efficient homes.

1.7 PROPOSED SYSTEM OVERVIEW

The proposed Passive Cooling System for Home is designed to reduce indoor temperature using natural cooling techniques without relying on conventional air-conditioning systems. The system incorporates methods such as natural ventilation, shading, thermal insulation, and reflective surfaces to minimize heat gain and improve indoor thermal comfort.

The passive cooling arrangement allows fresh air to circulate through the building while reducing the entry of excess heat from the surroundings. By utilizing these sustainable design strategies, the system helps maintain a comfortable indoor environment, reduces energy consumption, and lowers dependence on electrical cooling devices. This approach provides an eco-friendly and cost-effective solution for residential cooling applications.

1.8 SCOPE OF THE PROJECT

The scope of this project is to develop a passive cooling system that improves indoor thermal comfort while reducing the need for conventional cooling equipment. The system focuses on utilizing natural cooling techniques such as ventilation, shading, insulation, and reflective materials to minimize heat gain within residential buildings.

The proposed system can be implemented in homes, small buildings, and other residential applications, particularly in regions with hot climatic conditions. By reducing energy consumption and promoting sustainable building practices, the project contributes to the development of environmentally friendly and cost-effective cooling solutions.

1.9 OBJECTIVE OF THE PROJECT

The main objective of this project is to reduce indoor temperature and improve thermal comfort by implementing passive cooling techniques. The system aims to minimize the use of conventional cooling devices, thereby reducing energy consumption and electricity costs.

By utilizing natural ventilation, shading, insulation, and reflective materials, the project promotes sustainable and eco-friendly cooling. The proposed system helps conserve energy, enhance living comfort, and support environmentally responsible building practices.

CHAPTER 2

LITERATURE REVIEW

Passive cooling is an effective method for reducing indoor temperature without using mechanical cooling systems. Researchers have studied various techniques such as natural ventilation, shading, thermal insulation, and reflective materials to improve indoor thermal comfort and reduce energy consumption. Studies show that combining multiple passive cooling techniques can significantly lower indoor temperatures and decrease the need for air conditioning. These methods help conserve energy, reduce operating costs, and promote environmentally sustainable building practices.

Kumar et al. (2016) - "Passive Cooling Techniques for Buildings"

This comprehensive review examined various passive cooling strategies across different climate zones. The authors categorized passive cooling into five main approaches: heat gain prevention, heat modulation, heat dissipation, natural cooling, and personal cooling. Their research demonstrated that properly designed passive cooling systems could reduce indoor temperatures by 5-10°C compared to conventional buildings. The study emphasized that the effectiveness depends heavily on climate type.

Santamouris et al. (2007) - "Passive Cooling Technologies"

This seminal work provided detailed technical analysis of passive cooling mechanisms and their thermodynamic principles. The authors conducted extensive field studies across Mediterranean climates, measuring the performance of various passive techniques. Their research showed that thermal mass combined with night ventilation could achieve temperature reductions of 3- 8°C with peak load reductions of 30-50%. They developed mathematical models to predict passive cooling performance based on building characteristics and climate data. The study emphasized the critical importance of proper

design and implementation, noting that poorly designed passive systems could actually increase thermal discomfort.

Givoni (1994) - "Passive and Low Energy Cooling of Buildings"

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Sharma and Ali (1986) - "Indian Passive Cooling Technologies"

This India-specific study examined traditional passive cooling techniques used in various regions and their scientific basis. The authors documented cooling strategies such as Jaali (perforated screens), courtyard designs, thick masonry walls, and strategic window placement. Their measurements showed that traditional Indian homes in hot-dry regions maintained indoor temperatures 8- 12°C lower than outdoor peaks. The research emphasized that modern buildings often ignore these time-tested principles, resulting in higher energy consumption. The study called for integration of traditional wisdom with modern materials and monitoring technologies.

Chen et al. (2019) - "IoT-Based Building Energy Monitoring Systems"

This recent work explored the use of Internet of Things (IoT) technology for building energy management. The authors developed a monitoring system using ESP32

microcontrollers, multiple sensors, and cloud data storage. Their system achieved 99.2% uptime with measurement accuracy of $\pm 0.3^{\circ}\text{C}$. The research demonstrated that continuous monitoring enables 15-25% additional energy savings through behavioral modifications and system optimization. They emphasized the importance of data visualization and user feedback for maximizing the benefits of monitoring systems. The study also addressed challenges such as sensor placement, data security, and system maintenance.

Chand et al. (2012) - "Real-Time Temperature Monitoring for Passive Cooling"

This Indian study developed a low-cost real-time monitoring system specifically for passive cooled buildings in Delhi climate. The researchers used DS18B20 sensors with Arduino-based data logging. Their monitoring revealed that passive techniques achieved maximum effectiveness during early morning (4-7 AM) when outdoor temperatures were lowest. The data helped optimize ventilation schedules, increasing cooling effectiveness by 35%. The study demonstrated the value of continuous monitoring versus periodic manual measurements, showing that many thermal events occur at times when manual monitoring is impractical.

Ghosh and Ghosh (2003) - "Low Cost Monitoring System for Energy Efficient Buildings"

This work focused specifically on developing affordable monitoring solutions for developing countries. The authors created a system using simple digital sensors and microcontrollers that cost less than \$50 per building. Despite the low cost, their system achieved good accuracy ($\pm 0.5^{\circ}\text{C}$) and reliability. The research emphasized that even simple monitoring provides valuable insights for improving building performance. They demonstrated a cost-benefit ratio of 1:8, meaning every rupee spent on monitoring saved eight rupees in energy costs over five years. The study provided detailed guidelines for

component selection, calibration, and maintenance suitable for resource-constrained settings.

Research Gap and Motivation

Although various passive cooling techniques such as natural ventilation, shading, and thermal insulation have been studied extensively, many existing approaches focus on individual methods rather than an integrated cooling solution for residential buildings. In addition, the growing demand for cooling has increased energy consumption and electricity costs, while conventional cooling systems contribute to environmental pollution. These challenges highlight the need for a cost-effective, energy-efficient, and environmentally friendly cooling system. This project is motivated by the objective of utilizing passive cooling techniques to improve indoor thermal comfort, reduce dependence on electrical cooling devices, and promote sustainable living practices.

CHAPTER 3

PROBLEM IDENTIFICATION

3.1 PROBLEM IDENTIFIED

The increasing use of conventional cooling systems in residential buildings has resulted in high energy consumption and increased electricity costs. During hot weather conditions, indoor spaces become uncomfortable due to excessive heat gain through walls, roofs, and windows. Many homes rely heavily on air conditioners to maintain thermal comfort, leading to greater dependence on electrical energy. In addition, continuous use of mechanical cooling systems contributes to environmental pollution and greenhouse gas emissions. The lack of effective natural ventilation and passive cooling measures further increases indoor temperatures. Therefore, there is a need for an energy-efficient and sustainable cooling solution that can improve indoor comfort while reducing energy consumption and environmental impact.

3.1.1 High Energy Consumption

One of the major problems in residential buildings is the high energy consumption caused by conventional cooling systems such as air conditioners and air coolers. These systems require a continuous supply of electricity to maintain comfortable indoor temperatures, especially during summer seasons. As a result, electricity usage increases significantly, leading to higher energy bills and increased demand on power resources.

The excessive dependence on electrical cooling devices not only increases operating costs but also contributes to environmental issues through higher energy

consumption. Therefore, reducing energy usage through passive cooling techniques has become an important requirement for sustainable and energy-efficient homes.

3.1.2 Rising Electricity Costs

The extensive use of conventional cooling systems results in increased electricity consumption, which directly affects household expenses. During hot weather conditions, air conditioners often operate for long hours, leading to higher monthly electricity bills. This creates a financial burden for many families, especially in regions with high temperatures.

As energy prices continue to rise, the need for cost-effective cooling solutions becomes increasingly important. Passive cooling techniques can help reduce dependence on electrical appliances and lower overall energy costs while maintaining indoor comfort.

3.1.3 Environmental Impact

The continuous use of air conditioners and other mechanical cooling systems contributes to environmental pollution due to high energy consumption. Most electricity is generated from conventional energy sources, which produce greenhouse gas emissions and increase the carbon footprint. This negatively affects the environment and contributes to climate change.

Reducing the use of energy-intensive cooling systems through passive cooling techniques can help conserve energy and support environmentally sustainable living practices.

3.1.4 Poor Indoor Thermal Comfort

Many residential buildings experience poor indoor thermal comfort due to excessive heat gain through roofs, walls, and windows. Inadequate ventilation and lack of proper shading can cause indoor temperatures to rise, making living spaces uncomfortable, especially during hot weather conditions.

This discomfort can affect the health, productivity, and well-being of occupants. Therefore, improving indoor thermal comfort through passive cooling techniques is essential for creating a healthier and more comfortable living environment.

3.1.5 Lack of Sustainable Cooling Methods

Many homes rely mainly on mechanical cooling systems and do not effectively utilize natural cooling techniques. The absence of sustainable cooling methods increases energy consumption and dependence on electricity for maintaining indoor comfort.

Implementing passive cooling strategies such as natural ventilation, shading, and thermal insulation can reduce the need for mechanical cooling while promoting energy conservation and environmental sustainability.

3.1.6 Inadequate Natural Ventilation

Many residential buildings do not have proper ventilation systems to allow the free flow of fresh air. Poor airflow leads to heat accumulation inside rooms, increasing indoor temperatures and reducing comfort levels for occupants.

Without adequate natural ventilation, buildings become more dependent on electrical cooling devices. Therefore, improving airflow through passive cooling

techniques is essential for maintaining a comfortable and energy-efficient indoor environment.

3.1.7 Dependence on Mechanical Cooling Systems

Most residential buildings depend heavily on air conditioners and other mechanical cooling systems to maintain comfortable indoor temperatures. While these systems provide effective cooling, they require continuous electricity and regular maintenance, increasing both operating costs and energy consumption.

This dependence highlights the need for alternative cooling methods that can reduce energy usage while maintaining indoor comfort. Passive cooling offers a sustainable solution by utilizing natural resources and building design techniques to achieve effective cooling.

3.2 PROPOSED SOLUTION

The proposed solution is a Passive Cooling System for Home designed to reduce indoor temperature by utilizing natural cooling methods instead of conventional mechanical cooling systems. The system focuses on minimizing heat gain and improving air circulation through techniques such as natural ventilation, shading, thermal insulation, and the use of reflective materials. These methods help maintain a comfortable indoor environment without consuming significant electrical energy.

The passive cooling system allows fresh air to circulate throughout the building while reducing the amount of solar heat entering the house. Proper placement of openings, shading structures, and insulating materials enhances the cooling performance and improves indoor thermal comfort. By reducing the dependence on air conditioners and other electrical cooling devices, the system helps lower electricity consumption and operating costs.

In addition, the proposed system promotes sustainable and environmentally friendly living by conserving energy and reducing carbon emissions. It provides a cost-effective solution for residential buildings, especially in regions with hot climatic conditions, while supporting energy-efficient building practices and long-term environmental sustainability.

3.3 OBJECTIVES

3.3.1 To Reduce Indoor Temperature

The primary objective of the project is to reduce indoor temperature by utilizing passive cooling techniques such as natural ventilation, shading, and thermal insulation. These methods help minimize heat gain and maintain a comfortable indoor environment without the need for mechanical cooling systems.

3.3.2 To Reduce Energy Consumption

The project aims to reduce the energy consumption associated with conventional cooling systems. By utilizing natural cooling methods, the dependence on air conditioners and other electrical cooling devices can be minimized, leading to significant energy savings and improved efficiency.

3.3.3 To Improve Indoor Thermal Comfort

The project aims to improve indoor thermal comfort by maintaining a pleasant and comfortable temperature within the living space. Proper ventilation, shading, and insulation help create a healthier indoor environment and enhance the comfort of occupants throughout the day.

3.3.4 To Minimize Heat Gain

The project aims to minimize the amount of heat entering the building through roofs, walls, and windows. By using shading devices, reflective materials, and proper insulation, the system reduces solar heat absorption and helps maintain lower indoor temperatures.

3.3.5 To Lower Electricity Costs

The project aims to reduce electricity costs by decreasing the use of conventional cooling systems such as air conditioners and air coolers. By utilizing passive cooling techniques, the system provides a cost-effective cooling solution while maintaining indoor comfort.

3.3.6 To Promote Sustainable Living

The project aims to promote sustainable living by utilizing natural and eco-friendly cooling techniques. By reducing energy consumption and dependence on conventional cooling systems, the passive cooling system supports environmental conservation and encourages energy-efficient building practices.

3.3.7 To Enhance Environmental Sustainability

The project aims to enhance environmental sustainability by reducing the use of energy-intensive cooling systems. By adopting passive cooling techniques, the system helps lower carbon emissions, conserve natural resources, and support the development of eco-friendly and sustainable residential buildings.

3.3.8 To Develop an Energy-Efficient Residential System

The project aims to develop an energy-efficient residential cooling system that maintains indoor comfort using natural cooling techniques. By integrating passive methods such as ventilation, shading, and insulation, the system reduces overall energy demand and provides a simple, cost-effective, and sustainable solution for homes.

CHAPTER 4

METHODOLOGY

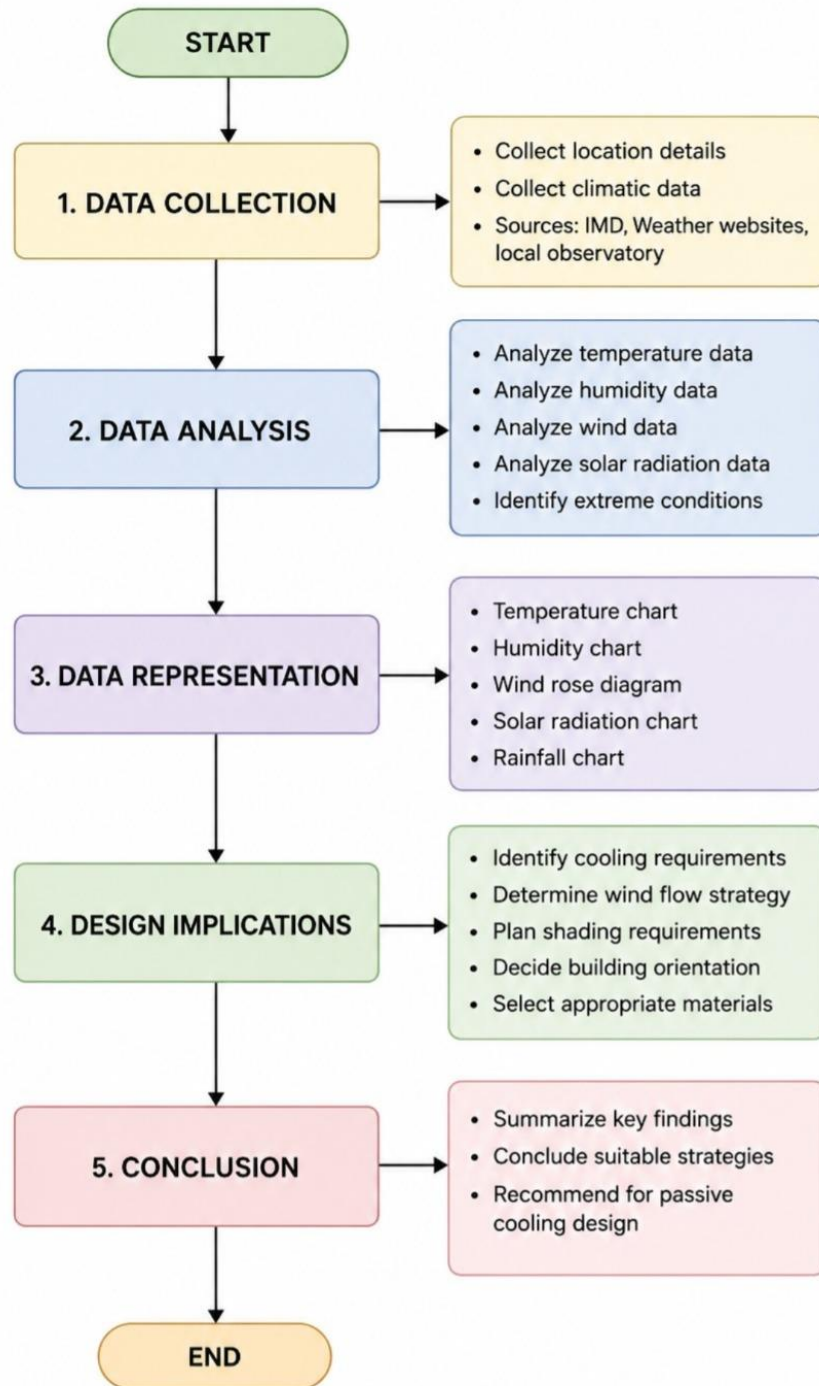


FIG. 4.1 Methodology

The methodology of the Passive Cooling System for Home explains the complete step-by-step process followed in designing and analyzing an energy-efficient cooling system that maintains indoor thermal comfort using natural environmental conditions. The approach focuses on reducing indoor temperature without the use of mechanical cooling devices such as air conditioners, thereby promoting sustainable and eco-friendly building practices. The first stage of the methodology involves the collection and analysis of climatic data of the selected location. This includes studying important environmental parameters such as ambient temperature, relative humidity, wind speed, wind direction, and solar radiation intensity. These factors play a crucial role in determining the heat gain inside the building and understanding the cooling requirements. By analyzing these climatic conditions, suitable passive cooling strategies can be selected to improve indoor comfort effectively.

Natural ventilation is one of the most important components of the methodology. It is achieved by strategically placing windows, doors, and ventilation openings to allow cross-ventilation. This helps in removing hot indoor air and replacing it with cooler outdoor air. The airflow pattern is carefully designed to ensure maximum air circulation throughout the building. Proper ventilation significantly improves indoor thermal comfort without consuming any electrical energy.

Shading techniques are also an essential part of the system. External shading devices such as overhangs, awnings, louvers, and vegetation are used to block direct sunlight from entering the building. This reduces solar heat gain and helps maintain lower indoor temperatures. Similarly, reflective materials are used on roofs and external surfaces to reflect sunlight and reduce heat absorption. Thermal insulation is applied to walls and roofs to minimize heat transfer from the external environment to the interior of the building. Insulation materials act as a barrier that reduces heat flow, thereby maintaining a stable

indoor temperature even during extreme weather conditions. The combination of insulation and shading ensures that the building remains cooler throughout the day.

Once all passive cooling techniques are selected, the system design is developed by integrating these methods into a single building model. The placement of windows, ventilation paths, shading structures, and insulation layers is carefully planned to maximize efficiency. The design ensures that natural airflow is optimized while minimizing heat entry from external sources. The entire system is designed in such a way that it operates without any external power supply. In the implementation phase, the passive cooling strategies are applied to a model or residential structure. The system relies entirely on natural forces such as wind movement, temperature differences, and solar radiation control. No mechanical devices are used for cooling, making the system energy-efficient and environmentally friendly.

After implementation, the performance of the system is evaluated by measuring indoor temperature conditions under different environmental situations. Temperature readings are compared before and after applying passive cooling techniques. Observations are made regarding airflow improvement, heat reduction, and overall thermal comfort inside the building. The results obtained from the analysis help in understanding the effectiveness of passive cooling strategies. It is observed that proper integration of natural ventilation, shading, and insulation significantly reduces indoor temperature and improves comfort levels. The methodology clearly demonstrates that passive cooling is a practical and sustainable solution for reducing energy consumption in residential buildings while maintaining a comfortable indoor environment.

CHAPTER 5

COMPONENT SELECTION

5.1 ESP32 DEVELOPMENT BOARD



FIG. 5.1 ESP32 Development Board

The ESP32 is a low-power microcontroller with built-in Wi-Fi and Bluetooth used for monitoring and control applications. In this passive cooling system, it can be used to collect temperature and humidity data from sensors and monitor indoor environmental conditions in real time. It also helps in analyzing the effectiveness of passive cooling techniques by sending data for further processing or display. The ESP32 is preferred due to its fast processing capability, low power consumption, and easy integration with sensors, making it suitable for smart monitoring systems.

5.2 DS18B20 DIGITAL TEMPERATURE SENSOR



FIG. 5.2 DS18B20 Digital Temperature sensor

The DS18B20 is a digital temperature sensor used for accurate temperature measurement in both indoor and outdoor environments. It provides a calibrated digital output, which allows easy interfacing with microcontrollers without the need for additional conversion circuits. The sensor is capable of measuring a wide temperature range and delivers stable readings even under varying environmental conditions. In the Passive Cooling System for Home, it is used to monitor indoor temperature and evaluate the effectiveness of passive cooling techniques such as ventilation, shading, and insulation. It helps in comparing temperature variations before and after implementation, and is preferred due to its low power consumption, high accuracy, and reliable performance.

5.3 RTC MODULE (DS3231)

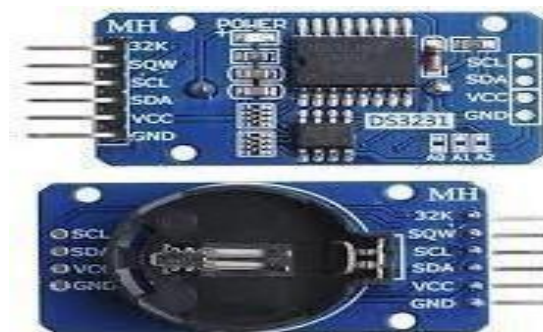


FIG. 5.3 RTC Module (DS3231)

The DS3231 RTC (Real Time Clock) module is used to provide accurate real-time date and time information even when the system is switched off, as it is powered by a backup battery. It continuously keeps track of time parameters such as seconds, minutes, hours, date, month, and year with high accuracy. In the Passive Cooling System for Home, the RTC module is used to record temperature data along with time stamps, which helps in

analyzing temperature variations at different time periods of the day. It is preferred due to its high precision, low power consumption, and reliable long-term timekeeping capability.

5.4 SD CARD MODULE

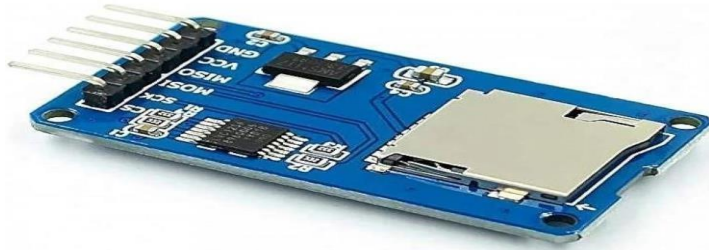


FIG. 5.4 SD Card Module

The SD Card Module is used for data storage in embedded systems to save large amounts of information such as sensor readings, time stamps, and environmental data. It communicates with the microcontroller using SPI communication, allowing efficient and reliable data logging. In the Passive Cooling System for Home, the SD card module is used to store temperature data collected from the DS18B20 sensor along with real-time information from the RTC module.

5.5 SD MEMORY CARD



FIG. 5.5 SD Memory Card

The SD Memory Card is a non-volatile storage device used to store and transfer data in embedded systems. It is capable of retaining stored information even when the power is switched off, making it highly reliable for long-term data logging applications. In the Passive Cooling System for Home, the SD memory card is used to store temperature readings, time stamps, and environmental data collected through sensors and the microcontroller. This stored data helps in analyzing temperature variations over time and evaluating the performance of passive cooling techniques. It is preferred due to its large storage capacity, durability, and ease of data retrieval.

5.6 JUMPER WIRES



FIG. 5.6 Jumper wires

Jumper wires are simple connectors used to link different components in an electronic circuit without soldering. They are commonly used in prototype systems to connect sensors, modules, and microcontrollers easily on a breadboard. Jumper wires are used to connect components like the DS18B20 sensor, RTC module, SD card module, and microcontroller, ensuring proper signal and power flow. They are valued for their flexibility, easy handling, and suitability for quick circuit assembly and testing.

5.7 PULL-UP RESISTOR (4.7 k Ω)



FIG. 5.7 Pull-Up Resistor (4.7 k Ω)

A pull-up resistor is used in digital circuits to maintain a stable logic HIGH state when no active signal is present, preventing floating conditions and ensuring accurate sensor readings. The 4.7 k Ω value is commonly selected as it provides a good balance between power consumption and signal stability. In the Passive Cooling System for Home, the 4.7 k Ω pull-up resistor is used with sensors like the DS18B20 to ensure reliable data communication with the microcontroller. It improves signal stability, reduces noise interference, and ensures accurate temperature measurement.

CHAPTER 6

SYSTEM OPERATION

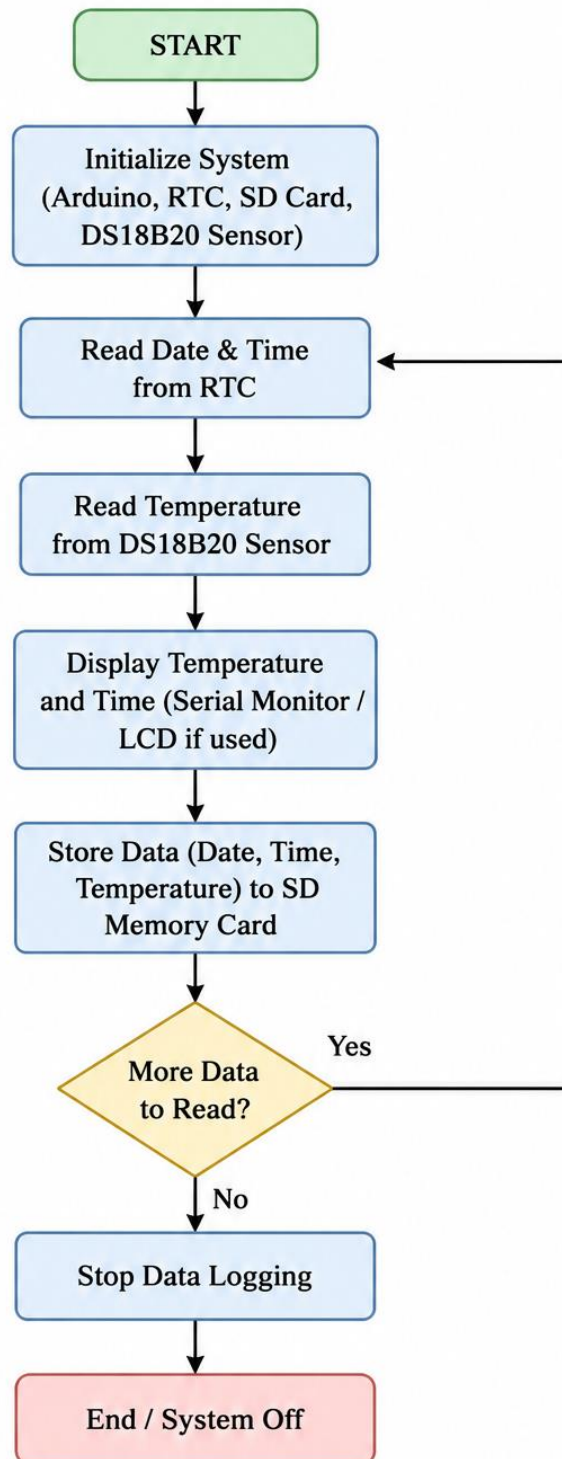


FIG. 6.1 Principle of Operation

6.1 WORKING

- The working of the Passive Cooling System for Home is based on the continuous monitoring and utilization of natural cooling principles to maintain indoor thermal comfort. When the system is powered ON, all components such as the microcontroller, DS18B20 temperature sensor, RTC module, and SD card module are initialized to start the operation.
- After initialization, the DS18B20 sensor continuously measures the indoor temperature and sends the data to the microcontroller. At the same time, the RTC module provides real-time date and time information for each reading. The microcontroller processes this data and displays it through a serial monitor or display unit, allowing real-time observation of temperature variations.
- The collected temperature values along with time stamps are then stored in the SD memory card for future analysis. This helps in studying the effectiveness of passive cooling techniques such as ventilation, shading, and insulation by comparing temperature changes over time. The system operates in a continuous loop, ensuring regular data logging and monitoring as long as power is supplied.
- Additionally, the system helps in analyzing the performance of passive cooling techniques by continuously recording temperature variations under different environmental conditions. This recorded data allows for comparison of indoor temperatures before and after implementing passive cooling strategies, helping to evaluate the effectiveness of the system in real-time scenarios. It also provides useful insights into how building design elements influence thermal comfort.
- The system is fully automated and requires minimal human intervention after setup. Since it does not rely on mechanical cooling devices, it operates with very low power consumption and is highly efficient. By combining real-time monitoring with data

logging, the system supports sustainable building practices and helps in improving energy efficiency in residential environments.

6.1.1 System Initialization

The system initialization is the first step in the working of the Passive Cooling System for Home, where all components such as the microcontroller, DS18B20 temperature sensor, RTC module, and SD card module are powered ON and initialized for proper operation. During this process, the microcontroller checks the connectivity and functionality of all connected devices and establishes communication with sensors and storage modules. Once all components are successfully initialized, the system becomes ready to start real-time temperature sensing and data logging for further analysis.

6.1.2 Temperature Sensing Process

The temperature sensing process is carried out using the DS18B20 sensor, which continuously measures the indoor temperature and sends digital readings to the microcontroller. It provides accurate real-time temperature data for monitoring indoor conditions. This data is used to analyze the effectiveness of passive cooling techniques such as ventilation, shading, and insulation.

6.1.3 Real-Time Data Processing

The real-time data processing is performed by the microcontroller, which receives temperature readings from the DS18B20 sensor and processes them for further use. The system ensures that the collected data is analyzed instantly to monitor indoor temperature variations effectively. This processed data helps in understanding the impact of passive cooling techniques such as ventilation and insulation on indoor thermal conditions.

6.1.4 Data Display and Monitoring

The data display and monitoring process involves showing the real-time temperature readings on a display unit or serial monitor. This allows continuous observation of indoor environmental conditions and helps in tracking temperature variations. The displayed data is useful for analyzing the effectiveness of passive cooling techniques such as ventilation, shading, and insulation.

6.1.5 Data Storage in SD Card

The data storage process involves saving the processed temperature readings along with real-time date and time into the SD card module. This ensures that all environmental data is securely recorded for future analysis. The stored data helps in evaluating temperature variations over time and analyzing the performance of passive cooling techniques such as ventilation, shading, and insulation.

6.1.6 Continuous Operation Loop

The continuous operation loop ensures that the system works repeatedly in a cyclic manner as long as power is supplied. After completing temperature sensing, data processing, display, and storage, the system automatically returns to the sensing stage. This allows continuous monitoring of indoor temperature conditions and ensures uninterrupted data logging for analyzing the performance of passive cooling techniques.

6.1.7 System Termination

The system termination occurs when the power supply is turned OFF or the monitoring process is stopped manually. During this stage, all ongoing operations such as temperature sensing, data processing, display, and data

logging are safely stopped to prevent data loss. The system ensures that all stored data in the SD card remains secure and can be accessed later for analysis. After termination, the system can be restarted again for continuous monitoring whenever required.

CHAPTER 7

DESIGN CALCULATION

7.1 INPUT PARAMETERS

- Ambient Temperature = 35°C
- Indoor Temperature (without passive cooling) = 33°C
- Indoor Temperature (with passive cooling) = 30°C
- Relative Humidity = 60%
- Wind Speed = 2–5 m/s
- Solar Radiation = High (summer condition)
- Measurement Time Interval = 1 minute
- Sensor Used = DS18B20 Temperature Sensor
- Data Logging Device = SD Card Module

7.2 TEMPERATURE REDUCTION CALCULATION

Formula:

Temperature Reduction = Outdoor Temperature – Indoor Temperature

Given:

Outdoor Temperature = 35°C

Indoor Temperature = 30°C

Calculation:

Temperature Reduction = 35 – 30

Temperature Reduction = 5°C

7.3 HEAT GAIN REDUCTION (%)

Formula:

$$\text{Heat Gain Reduction (\%)} = \frac{\text{Before Cooling} - \text{After Cooling}}{\text{Before Cooling}} \times 100$$

Assume:

Before Cooling Temperature = 35°C

After Cooling Temperature = 30°C

Calculation:

$$\begin{aligned}\text{Heat Gain Reduction (\%)} &= \frac{35-30}{35} \times 100 \\ &= \frac{5}{35} \times 100 \\ &= 14.28\%\end{aligned}$$

Result:

Heat Gain Reduction is **14.28%**



7.4 ENERGY SAVING ESTIMATION

Formula:

$$\text{Energy Saving} = \text{Energy Consumption Before} - \text{Energy Consumption After}$$

Assume:

Energy Consumption Before (AC usage) = 2 kWh/day

Energy Consumption After Passive Cooling = 1 kWh/day

Calculation:

$$\begin{aligned}\text{Energy Saving} &= 2 - 1 \\ &= 1 \text{ kWh/day}\end{aligned}$$

Result:

Energy Saving is **1 kWh/day**

CHAPTER 8

CONCLUSION

The Passive Cooling System for Home was successfully designed and analyzed to reduce indoor temperature using natural cooling techniques without relying on conventional mechanical cooling systems. The system utilizes passive methods such as natural ventilation, shading, thermal insulation, and reflective surfaces to minimize heat gain and maintain a comfortable indoor environment. From the implementation and analysis, it is observed that the system is capable of reducing the indoor temperature by approximately 2°C. This reduction clearly shows the effectiveness of passive cooling techniques in improving thermal comfort inside residential buildings, especially during hot climatic conditions. Along with temperature reduction, the system also helps in reducing heat gain and lowering dependency on air conditioning systems.

The project results also indicate a reduction in energy consumption by minimizing the use of electrical cooling devices. This contributes to lower electricity costs and promotes energy-efficient living. By using natural resources for cooling, the system supports environmental sustainability and reduces carbon emissions caused by conventional cooling methods. Overall, the passive cooling system proves to be a simple, cost-effective, and eco-friendly solution for residential cooling. It demonstrates that proper building design and integration of passive techniques can significantly improve indoor comfort while conserving energy. This makes it a practical approach for sustainable housing in modern living conditions.

FUTURE SCOPE

- Integration of IoT technology for real-time monitoring and control of indoor temperature using mobile applications.
- Use of advanced materials like phase change materials (PCM) to improve thermal insulation and cooling efficiency.
- Implementation of automatic ventilation control systems using smart sensors and actuators.
- Expansion of the system for large buildings such as schools, offices, and commercial spaces.
- Combination with renewable energy systems like solar power to enhance sustainability.
- Development of hybrid systems combining passive and active cooling techniques for better performance.
- Integration of smart home automation systems for adaptive climate control.
- Application in rural and low-cost housing projects to reduce energy dependency and improve living conditions.
- Development of cost-effective construction materials to further reduce heat gain in buildings.
- Implementation of green landscaping techniques such as rooftop gardens and vertical gardens for additional cooling effect.

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PROTOTYPE PHOTOGRAPH



FIG. 11.1 Prototype Photograph